SIS 1 Report: Exchange rates and national economies

Name	ID	Part
Dauletkhan Qian	23B031082	1. Data from the World Bank API
Kalybay Nurdaulet	23B031515	2. Web scraping GDP from Wikipedia
Dauirkhankyzy Balnur	23B031274	3. Cleaning and merging
Abibulla Kundyz	23B030352	4. Analysis and plots

1. Goal

Some currencies stay stable for years, others lose almost all their value. We wanted to know **if the size of an economy is connected with this**. Our idea was simple: a big economy (high GDP) should have a stronger and more stable currency. We also checked inflation, because it is the most common explanation why a currency gets weaker.

Period: 2015-2025. **Currency change** is measured against the US dollar. **Size of the economy** is nominal GDP in US dollars.

2. Data sources

Source	How we got it	What we took
World Bank API	requests, JSON, page by page	exchange rate (PA.NUS.FCRF, local currency per 1 USD), inflation (FP.CPI.TOTL.ZG), region and income group of every country
Wikipedia: List of countries by GDP (nominal)	requests + BeautifulSoup (robots.txt allows it)	GDP in millions of USD from three sources: IMF, World Bank and UN

Every member worked in a separate notebook (01-04). The notebooks pass data to each other through CSV files in the *data/* folder.

3. Part 1: World Bank API (Dauletkhan Qian)

The API gives data in pages, so we wrote a function that downloads all pages one by one. We got 296 entries in the country list and 5,830 rows of data (country + year + indicator).

What we found when we looked at the data:

- 79 of the 296 entries are not countries but groups like "World" or "Euro area". 217 are real countries.
- The exchange rate is available for about 95% of countries every year, inflation for about 85%. 2025 is not complete yet (185 countries have a rate).
- 5 countries have no exchange rate at all: closed economies (North Korea, Turkmenistan, Cuba) or places that use another country's currency (Palestine, British Virgin Islands).
- Example: 1 USD was about 222 tenge in 2015 and about 521 tenge in 2025.

4. Part 2: Web scraping (Kalybay Nurdaulet)

First we checked robots.txt, then downloaded the page and took the first table (222 rows). The data was "dirty": numbers are text with commas, names have footnotes ("China [n 1]"), missing values are written as "N/a", some numbers have an older year in brackets, and the first row is the whole world.

We also checked if the three sources agree. For 196 countries that have both IMF and World Bank values the median difference is only about 9% (IMF is a newer year). Big differences are only in countries with war or crisis (Syria, Yemen, South Sudan, Lebanon). So the sources are reliable, and we used IMF first. The top 10 economies make about 66% of world GDP, so GDP is very uneven and we used log10(GDP) later.

5. Part 3: Cleaning and merging (Dauirkhankyzy Balnur)

Exchange rates. We removed groups, countries that use the US dollar (the rate is always 1) and Zimbabwe (changed its currency). For Croatia we converted the old kuna rates to euro. We kept only countries with data for 2015, 2025 and at least 9 of 11 years, and filled one small gap with interpolation. How many countries were left after each step:

Step	Countries
in World Bank data	217
has some exchange rate data	212
without US dollar countries	196
without currency change (Zimbabwe)	195
has 2015, 2025 and 9+ years	167
merged with GDP	166

New columns for every country: *value change* of the currency against USD from 2015 to 2025 (negative = weaker), *volatility* (how much it jumps from year to year), the *worst year* and *average inflation*.

GDP. We removed footnotes, brackets and commas, turned "N/a" into empty values and removed the "World" row. We took IMF GDP, and World Bank or UN if IMF was missing.

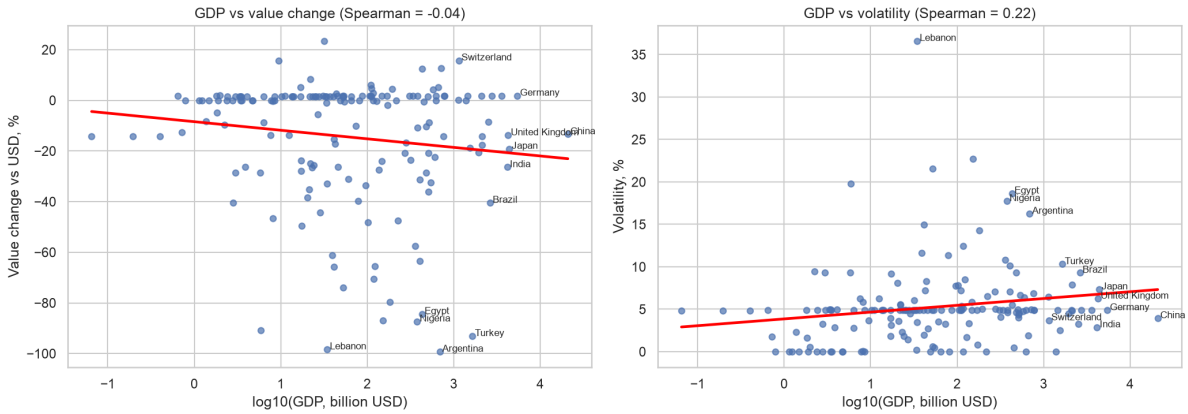
Merging. We merged by country name. 37 names were written differently ("Korea, Rep." vs "South Korea"), so we fixed 36 of them with a dictionary. Only Gibraltar was lost. The final table has **166 countries**. Many countries share one currency (euro, CFA franc), so we also made a table with one row per currency (99 rows) to check the results. At the end we check the table with assert (no duplicates, no missing values, all rates above 0).

6. Part 4: Analysis (Abibulla Kundyz)

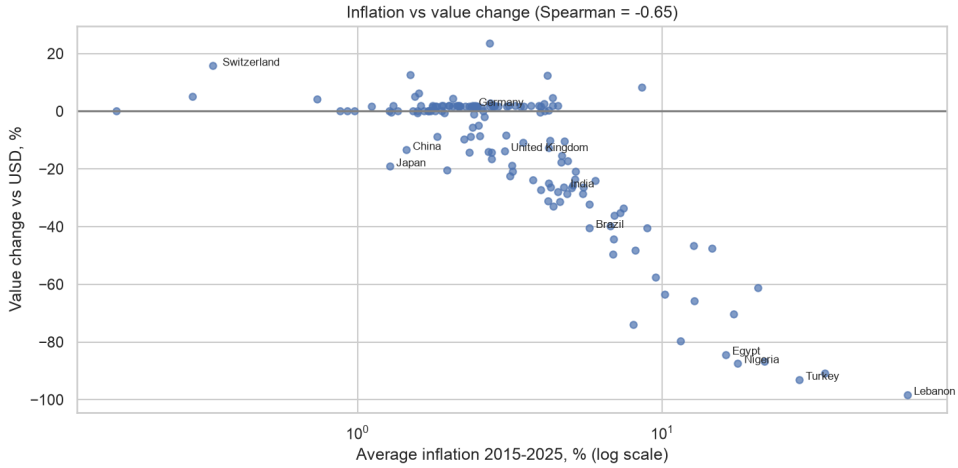
81 of 166 currencies (49%) became weaker against the dollar. Most of the others did not change because they are fixed to the dollar (Gulf countries) or to the euro (CFA franc, Denmark, Bulgaria). The biggest losses: Argentina -99%, Lebanon -98%, Turkey -93%, Suriname -91%, Nigeria -87%. Median change by income group: high income +1.6%, upper middle -13%, lower middle -26%, low income +1.6% (most of them use the CFA franc).

We used Spearman correlation, because there are big outliers and it works with ranks. The p-value shows if the connection can be just random ($p < 0.05$ means it is not random). "Floating only" means we removed currencies fixed to the dollar or euro (142 countries left).

Spearman correlation	all countries	p-value	one row per currency	floating only
log GDP and value change	-0.04	0.58	0.03	-0.02 (p = 0.85)
log GDP and volatility	0.22	0.005	0.14	0.09 (p = 0.27)
inflation and value change	-0.65	< 0.001		-0.74 (p < 0.001)
inflation and volatility	0.52	< 0.001		0.37 (p < 0.001)



GDP vs value change (left) and GDP vs volatility (right). The line on the left is almost flat.



Inflation vs value change: the higher the inflation, the more the currency lost.

7. Main findings

- **The size of the economy does not explain how much a currency lost.** The correlation is about 0 and not significant (p = 0.58). Big economies like Turkey, Brazil and India also lost a lot.
- Big economies looked a bit more volatile (0.22). But this is only because many small countries fix their currency. Without fixed currencies the connection disappears (0.09, p = 0.27). So our idea "bigger economy = more stable currency" was **not confirmed**.
- **Inflation is the main factor:** countries with high inflation lost much more (-0.65, and -0.74 for floating currencies).
- If we count every currency only once, the results are the same, so the euro countries did not change the result.

8. Limitations

- We used only yearly data, so we do not see changes inside a year.
- GDP is for one year, but exchange rates are for 11 years.
- Some countries were removed, and 20 countries have no inflation data (for example Argentina).
- Correlation does not mean causation: inflation and a weak currency can both be caused by other problems.